

Macroeconomic Effects of Rare Earths Supply Chain Disruptions

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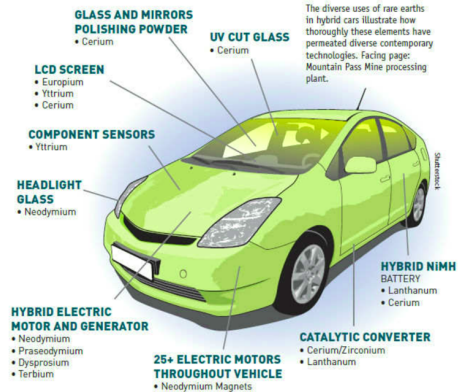
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Motivation

- **April 4, 2025 changed the risk calculus.** Shortly after new US tariffs, China introduced export licensing for seven REEs and related permanent magnets.
- The disruption was serious but short-lived. Yet it raised policymakers' concerns about a **prolonged, severe shortage.**



Source: minesmagazine.com/1737

What are the macroeconomic impact —output lost —of a major, prolonged disruption of REE supply across sectors and major economies?

Paper in a Nutshell

- **Measure exposure.** Combine sectoral REE use with input–output data for six economies.
- **Model propagation.** A small open economy with production networks and a binding REE import quota.
- **Quantify losses.** Simulate an 80% supply reduction under alternative substitution elasticities.

Contribution: connect direct REE dependence, production linkages, substitutability, and endogenous scarcity wedges

Direct Exposure to REEs Across Countries

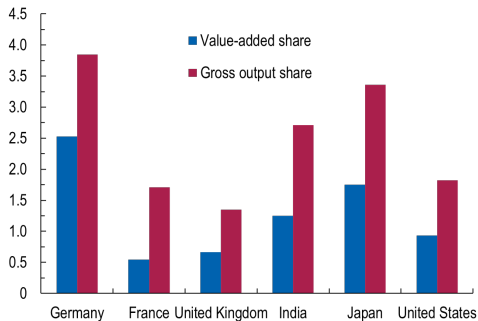


Figure: Value added and Gross Output at Risk
(OECD Input-Output Tables. IMF staff calculations.)

- For example, in the motor vehicle sector, if a truck valued at USD 100,000 uses a rare-earth catalyst, the full value of the truck is counted as exposed
- *It omits substitution, reallocation, and propagation through networks.*

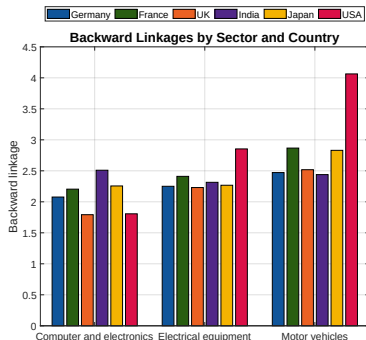
Small REEs-Magnets Input Shares (OECD Classification)

Electrical equipments, motor vehicles, and computer electronic are the sectors with the largest intermediate input share of REEs, with a share of 0.048%, 0.046%, and 0.044%, respectively.

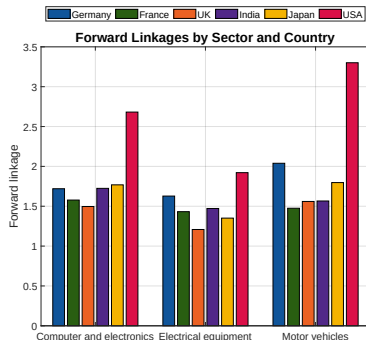
Sector OECD data	REEs	Magnets	Total
Electrical equipment	0.043	0.005	0.048
Motor vehicles, trailers and semi-trailers	0.002	0.044	0.046
Computer, electronic and optical equipment	0.022	0.022	0.044
Manufacture of other transport equipment	0.002	0.021	0.023
Manufacture of basic iron and steel	0.013	0.000	0.013
Machinery and equipment, nec	0.002	0.007	0.009
Other non-metallic mineral products	0.006	0.000	0.006
Fabricated metal products	0.003	0.000	0.003
Coke and refined petroleum products	0.003	0.000	0.003
Chemical and chemical products	0.003	0.000	0.003

Linkages of REE-Intensive Sectors

A backward (forward) linkage of 2.5 means a one-dollar of final demand for that (in each) sector requires \$2.50 of gross production (from this supplier) across the economy



(a) Backward Linkages



(b) Forward Linkages

Limited substitutability of REEs

Substitutes often exist, but at a cost in performance or after months/years of investment/innovation (Alfaro et al., 2025).

How difficult is substitution?

Substitutability index: higher means harder
100 = no adequate substitute

Heavy REEs 78



Non-REE elements 57



Graedel et al. (2015)

What limits substitution?

- **Performance penalties:** lower efficiency, greater weight or size, or higher costs.
- **Critical applications:** particularly limited options for HREEs such as dysprosium in permanent magnets.
- **Time and investment:** developing alternatives, including motors without REE permanent magnets, requires both.

Implication: scope for substitution is especially limited in the short run.

Theoretical Framework

- Dynamic multi-sector open economy: $N + 1$ domestic production sectors.
 - Borrowing-Saving in one-period foreign bond
 - Inelastic labour supply.
 - Labour, domestic and imported intermediate inputs.
 - Domestic input-output linkages
 - Non-unitary elasticities in production
- Quantity constraint on imported intermediates (REEs). Import Constraint

Households

Production

Intermediate Inputs

Exogenous Processes

Market Clearing

A Binding Quantity Constraint: Scarcity Wedge

Available REEs constrain total domestic use:

$$\sum_{i=1}^{N+1} M_{iM,t} = \bar{M}_{M,t}.$$

A common shadow value raises the effective input price:

$$P_{M,t}^{\text{eff}} = P_M + \mu_t = P_M(1 + \tau_t), \quad \tau_t = \frac{\mu_t}{P_M}.$$

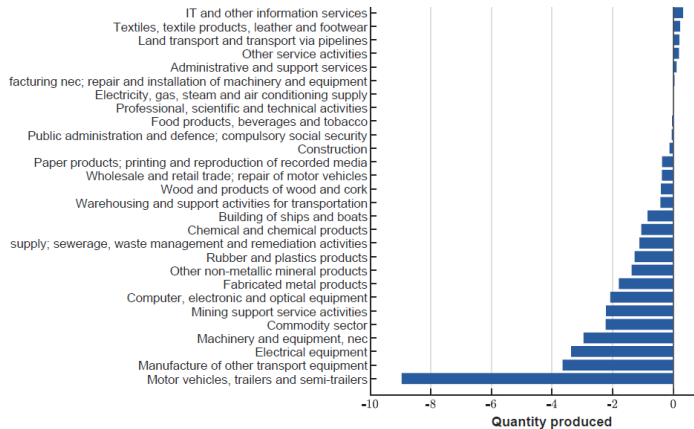
- The wedge adjusts endogenously to clear the REE market.
- Scarcity-related payments leave the domestic economy and are not rebated to households income.

Calibration and Experiment

- **Countries:** U.S., Germany, France, UK, Japan, and India.
- **Data:** OECD 2017 networks and factor shares; USGS REE-use shares, common across countries within each sector.
- **Shock:** 80% lower REE supply on impact; persistence $\rho_M = 0.9$.
- **Initial condition:** Slightly binding constraint $\sum_{i=1}^{N+1} = M_{iM,t} = 3.69 \leq \overline{M}_{M,t} = 3.67$

Scenario	σ	ϵ^D	ϵ
Low substitution	0.015	0.015	0.015
Only REE substitution low	0.8	0.8	0.015
High substitution	0.8	0.8	0.8

U.S. Sectoral Effects: Losses and Reallocation

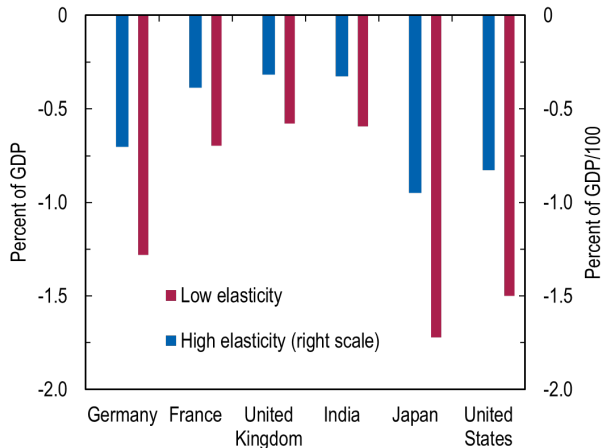


Low substitution

- Motor vehicles: -8.96% .
- Output falls in 19 sectors.
- Eighteen sectors expand slightly.

Impact responses to an 80% REE supply reduction.

GDP Losses in Import-Price Units



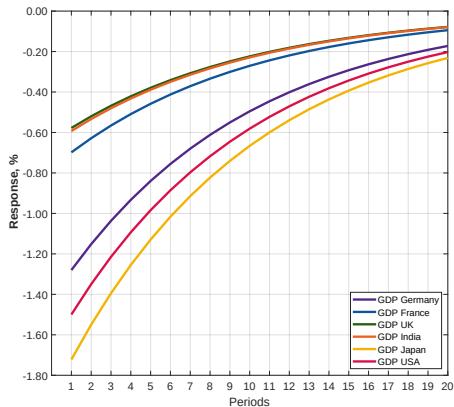
Low substitution

- Japan: -1.8% .
- U.S.: -1.5% .
- Germany: -1.2% .

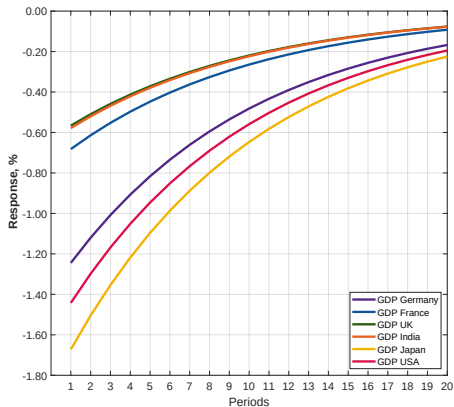
High substitution makes losses negligible.

Source: Impact responses. Nominal GDP measured relative to P_M .

Which Elasticity Drives Aggregate Losses?



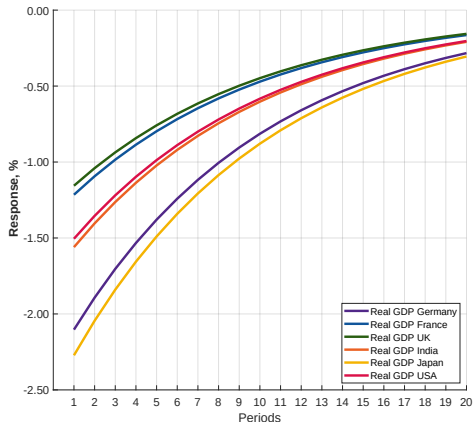
(a) Low Elasticity ($\sigma = \epsilon_D = \epsilon = 0.015$)



(b) Low-High Elasticity ($\sigma = \epsilon^D = 0.8$,
 $\epsilon = 0.015$)

Keeping only REE substitution low preserves 96–98% of the baseline GDP response.

Real GDP: The Country Ranking Changes



Removing the domestic GDP deflator

- Japan: -2.3% .
- Germany: -2.1% .
- U.S.: -1.5% .

Sector size and intermediate intensity explain Germany's greater exposure.

Conclusions

Small input shares can conceal large macroeconomic risks.

- **Networks transmit shortages** to suppliers and customers.
- **Economic structure shapes losses:** sector size, input intensity, and linkages.
- **REE substitution is decisive:** greater flexibility reduces scarcity wedges and GDP losses.

Policy implication: assess critical dependencies beyond direct import exposure.

Annex

Intertemporal optimization problem:

$$\begin{aligned} \max_{\{C_t, B_t\}_{t=0}^{\infty}} \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\rho} - 1}{1-\rho} \\ \text{s.t.} \quad & P_t C_t + P_{N+1,t}(B_t + g(B_t)) \leq W_t \bar{L} + (1+r)P_{N+1,t}B_{t-1} \\ & \text{given } B_{-1} \end{aligned}$$

- Adjustment costs of asset holding: $g(B_t) = \frac{\chi}{2}(B_t - \bar{B})^2$
- Cost intensity $\chi > 0$.
- Steady state level of debt $\bar{B}$.
- (Nominal) wage rate W_t .
- Labour supply $\bar{L}$.

Each sector produces output using labour and a composite of intermediate inputs:

$$\frac{Q_i}{\bar{Q}_i} = Z_i \left(\alpha_i \left(\frac{L_i}{\bar{L}_i} \right)^{\frac{\sigma_i - 1}{\sigma_i}} + (1 - \alpha_i) \left(\frac{M_i}{\bar{M}_i} \right)^{\frac{\sigma_i - 1}{\sigma_i}} \right)^{\frac{\sigma_i}{\sigma_i - 1}}$$

- Labour share: α_i ; share of intermediate inputs: $1 - \alpha_i$.
- The elasticity of substitution between labor and intermediates: σ_i .
- Intermediate inputs include both domestic and imported components.

The intermediate input bundle M_i :

$$\frac{M_i}{\bar{M}_i} = \left(\omega_i^D \left(\frac{M_i^D}{\bar{M}_i^D} \right)^{\frac{\epsilon_i - 1}{\epsilon_i}} + (1 - \omega_i^D) \left(\frac{M_{iM}}{\bar{M}_{iM}} \right)^{\frac{\epsilon_i - 1}{\epsilon_i}} \right)^{\frac{\epsilon_i}{\epsilon_i - 1}}$$

- Expenditure share on domestic (ω_i^D) and imported ($(1 - \omega_i^D)$) intermediates .
- Elasticity of substitution between domestic (M_i^D) and imported (M_{iM}) inputs: ϵ_i .

The domestic intermediate bundle:

$$\frac{M_i^D}{\bar{M}_i^D} = \left(\sum_{j=1}^{N+1} \omega_{ij} \left(\frac{M_{ij}}{\bar{M}_{ij}} \right)^{\frac{\epsilon_i^D - 1}{\epsilon_i^D}} \right)^{\frac{\epsilon_i^D}{\epsilon_i^D - 1}}$$

- Expenditure share on good i within the domestic intermediate bundle: ω_{ij} .
- Elasticity of substitution among domestic intermediates: ϵ_i^D .

Two key exogenous processes:

- The evolution of the commodity price in foreign currency:

$$\log P_{N+1,t}^* = \rho_{N+1} \log P_{N+1,t-1}^* + \varepsilon_{N+1,t}$$

- Sector-specific productivity:

$$\log Z_{i,t} = \rho_Z \log Z_{i,t-1} + \varepsilon_{i,t}$$

- Supply of imported REEs

$$\log \bar{M}_{M,t} = (1 - \rho_M) \log \bar{M} + \rho_M \log \bar{M}_{M,t-1} + \varepsilon_{M,t}$$

- Persistence parameters: ρ_{N+1} , ρ_Z , ρ_N

- i.i.d. shocks: $\varepsilon_{N+1,t}$, $\varepsilon_{i,t}$, $\varepsilon_{M,t}$.

- Output in each sector is fully absorbed by domestic final consumption and intermediate input demand:

$$Q_{i,t} = C_{i,t} + \sum_{j=1}^{N+1} M_{ji,t}, \quad \text{for } i = 1, 2, \dots, N$$

- Labour supply is fixed and allocated inelastically across sectors: $\bar{L} = \sum_{i=1}^{N+1} L_{i,t}$

- The evolution of the foreign asset position:

$$B_t = (1 + r)B_{t-1} - g(B_t) + X_t - \underbrace{\frac{P_{M,t}}{P_{N+1,t}} \left(\sum_{i=1}^{N+1} M_{iM,t} + C_{M,t} \right)}_{\text{Trade Balance}}$$

REE supply constraints

Import Constraint

- **Binding aggregate quota.** Domestic use of imported REEs satisfies

$$\sum_{i=1}^{N+1} M_{iM,t} = \bar{M}_{M,t}.$$

Supply follows an exogenous process. Its steady state is set slightly below unconstrained demand.

- **Common scarcity wedge.** The quota's shadow value $\mu_t \geq 0$ raises the effective input price in every sector:

$$\lambda_{i,t} \frac{\partial F_i}{\partial M_{iM,t}} = \underbrace{P_M + \mu_t}_{p_{M,t}^{\text{eff}}} = P_M(1 + \tau_t), \quad \tau_t \equiv \frac{\mu_t}{P_M}.$$

The wedge τ_t adjusts endogenously to clear the REE market.

- **Quota rents accrue abroad.** We assume that scarcity payments leave the domestic economy:

$$R_t^\tau = \tau_t P_M \bar{M}_{M,t}.$$

These payments are not rebated to domestic households.

The household minimizes the cost of allocating consumption across goods:

$$PC = \min_{\{C_i\}_{i=1}^{N+1}, C_M} \left\{ \sum_{i=1}^{N+1} P_i C_i + P_M C_M \right\}$$

s.t. the household consumption bundle:

$$\left(\frac{C_M}{\beta_M} \right)^{\beta_M} \cdot \prod_{i=1}^{N+1} \left(\frac{C_i}{\beta_i} \right)^{\beta_i} \geq C$$

- Consumption shares β_i satisfying $\sum_{i=1}^{N+1} \beta_i + \beta_M = 1$.
- $N + 1$ domestic sectors of production.
- The bundle of imported goods M .